

SAT Math Mastery · Week 7A

Scatterplots, Two-Way Tables & Probability

Time	150 minutes: 110 min solving + 40 min self-check and error log
Topics	line of best fit · predictions, interpolation vs extrapolation · residuals · slope and intercept in context · two-way tables · joint, marginal and conditional probability · independence · basic probability and expected counts
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Read a scatterplot and choose the equation of a line of best fit from its slope sign, size and intercept.
- S2.** Predict with a fitted line; tell interpolation (inside the data range) from extrapolation (outside it).
- S3.** Compute and interpret residuals: residual = actual – predicted.
- S4.** Interpret slope and intercept in context; rescale the slope for other steps of x and other units.
- S5.** Read two-way tables: joint, marginal and conditional counts, with the correct denominator.
- S6.** Conditional probability, including “given not in a group” and “given at least” (sum the remaining columns).
- S7.** Solve for a missing table entry from a stated probability or from independence.
- S8.** Simple and combined probability from counts, without replacement, and expected counts for a population.

2 · REQUIRED FACTS

- The slope of the line of best fit is the *predicted change in y for each 1-unit increase in x* ; the y -intercept is the predicted y at $x = 0$ (meaningful only if $x = 0$ makes sense).
- A positive residual means the point lies *above* the line (the line under-predicts); a negative residual means it lies below.
- A least-squares line passes through $(\bar{x}, \bar{y})$ and its residuals add up to 0.
- Predicting inside the range of the data is interpolation; outside it is extrapolation (less reliable).
- In a two-way table the *given* group is the denominator; the target group within it is the numerator.
- Probabilities are between 0 and 1; the probabilities of all outcomes of one experiment add to 1.

3 · REQUIRED FORMULAS

$$\hat{y} = mx + b, \quad \text{residual} = y - \hat{y}, \quad \hat{y}(\bar{x}) = \bar{y}$$

$$P(A) = \frac{\text{favorable}}{\text{total}}, \quad P(A | B) = \frac{n(A \text{ and } B)}{n(B)}, \quad P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\text{independent} \iff P(A | B) = P(A), \quad \text{expected count} = \text{population} \times \text{sample proportion}$$

$$\text{without replacement (two reds): } \frac{r}{n} \cdot \frac{r-1}{n-1}$$

4 · MINI THEORY SHEET

Choose the line of best fit.

Look at	Decide
trend direction	positive slope if points rise, negative if they fall
two points the line crosses	$m = \frac{\Delta y}{\Delta x}$, then compare with the options
y -intercept	where the line meets $x = 0$ (check the axis starts at 0)
points above/below	about as many above as below, and the line stays close to all

Residuals. Plug the point's x in the line to get $\hat{y}$, then subtract: actual – predicted. On a graph: vertical gap, above the line is positive.

Units. Slope per 1 year becomes slope $\times 10$ per decade; if every y is multiplied by k , then both m and b are multiplied by k ; shifting x by c changes only the intercept.

Two-way tables. Read the question in the order “Of those who (given) ... what fraction (target)?” The first group gives the denominator. “Not in a group” or “at least” means add the other columns.

Joint vs conditional. Joint: divide by the grand total. Conditional: divide by the given group's total.

Percent tables. If a table shows percents of the whole, a conditional probability is one entry divided by a marginal, not by 100%.

Speed shortcuts.

Situation	Shortcut
which equation fits the plot	test the slope sign, then the y -intercept; eliminate before computing
residual sign	above the line = +, below = –; no calculation needed to compare
slope per k units of x	multiply the slope by k
prediction for x	plug in; check whether x is within the data range first
$\hat{y}$ unknown intercept	use $(\bar{x}, \bar{y})$: $b = \bar{y} - m\bar{x}$
missing cell with a given probability	write the conditional fraction = given value, cross-multiply
independence	compare two conditional fractions (cross-multiply: $ad = bc$)
“at least” / “not” groups	sum the remaining columns for the denominator
without replacement	numerator and denominator both drop by 1

5 · GUIDED EXAMPLES

Example 1 — Residual and prediction

A line of best fit is $\hat{y} = 2.5x + 8$. A data point is (6, 21). Find the predicted value and the residual, and say whether the point lies above or below the line.

$\hat{y} = 2.5(6) + 8 = 23$. Residual = $21 - 23 = -2$. Negative residual: the point lies *below* the line.

Example 2 — Conditional probability from a two-way table

	Tea	Coffee	Total
Under 30	18	42	60
30+	36	24	60
Total	54	66	120

Of those who chose coffee, what fraction are under 30? The *given* group is coffee drinkers (total 66): $\frac{42}{66} = \frac{7}{11}$.
(Not $\frac{42}{60}$ and not $\frac{42}{120}$.)

Example 3 — Missing cell from a stated probability

Among $x + 20$ people who prefer plan B, x are members. If $P(\text{member} \mid B) = \frac{3}{5}$, find x .

$$\frac{x}{x + 20} = \frac{3}{5} \Rightarrow 5x = 3x + 60 \Rightarrow x = 30.$$

6 · PRACTICE SET — 32 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

Students in grades 10 and 11 chose their lunch. The results are shown.

	Pizza	Salad	Total
Grade 10	48	32	80
Grade 11	54	66	120
Total	102	98	200

Of the students who chose salad, what fraction were in grade 11?

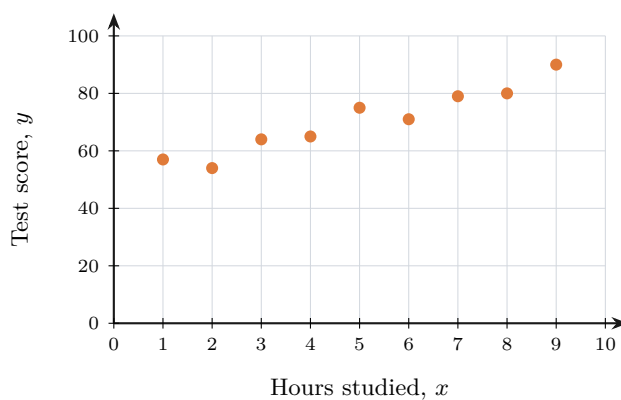
- (A) $\frac{16}{49}$
(C) $\frac{11}{20}$

- (B) $\frac{33}{49}$
(D) $\frac{33}{100}$

2.

EASY · DESMOS

The scatterplot shows the relationship between the number of hours x a student studied and the student's test score y . Which equation could represent the line of best fit?



(A) $y = -4x + 50$

(B) $y = 4x + 30$

(C) $y = 4x + 50$

(D) $y = 50x + 4$

3.

EASY · NO CALCULATOR NEEDED

A jar contains 12 red, 18 blue and 10 green marbles. One marble is chosen at random. What is the probability that the marble is not red?

4.

EASY · NO CALCULATOR NEEDED

The equation $\hat{y} = 3.2x + 14$ models the predicted height $\hat{y}$, in centimeters, of a sunflower x weeks after planting. Which is the best interpretation of the number 3.2 in this context?

(A) The predicted height increases by 3.2 cm each week. (B) The predicted height at planting is 3.2 cm.

(C) The predicted height after 14 weeks is 3.2 cm. (D) The predicted height increases by 14 cm each 3.2 weeks.

5.

EASY · CALCULATOR

The equation $\hat{y} = 26.4 - 1.8x$ models the predicted price $\hat{y}$, in thousands of dollars, of a used car that is x years old. According to the model, what is the predicted price, in thousands of dollars, of a 6-year-old car?

6.

MEDIUM · NO CALCULATOR NEEDED

Members of a community center were asked about their age group and preferred class.

	18–34	35–54	55+	Total
Yoga	34	46	40	120
Spin	52	38	10	100
Total	86	84	50	220

If one member is selected at random, what is the probability that the member is 35–54 years old and prefers spin?

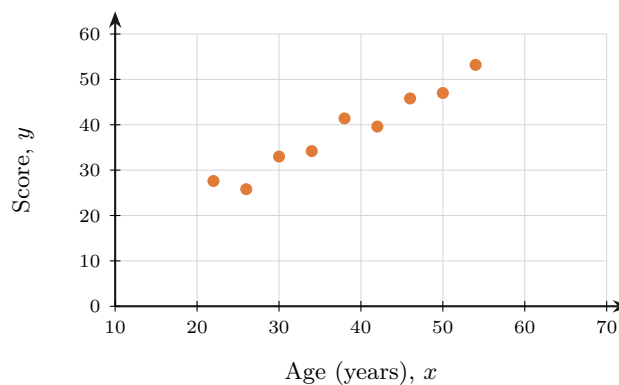
(A) $\frac{19}{50}$
(C) $\frac{19}{60}$

(B) $\frac{19}{42}$
(D) $\frac{19}{110}$

7.

MEDIUM · NO CALCULATOR NEEDED

The scatterplot shows data for people aged 22 to 54. A line of best fit is used to make predictions. Which of the following predictions would be an example of extrapolation?



(A) A prediction for a person aged 40

(B) A prediction for a person aged 50

(C) A prediction for a person aged 60

(D) A prediction for a person aged 25

8.

MEDIUM · NO CALCULATOR NEEDED

A line of best fit for a data set is $\hat{y} = 4.2x + 15$. For the data point with $x = 10$, the actual value is $y = 53$. What is the residual for this point? (residual = actual – predicted)

9.

MEDIUM · NO CALCULATOR NEEDED

The equation $\hat{y} = 0.84x + 1.5$ models the predicted height $\hat{y}$, in meters, of a tree x years after planting. According to the model, by how many meters is the tree's predicted height expected to increase every 5 years?

(A) 0.168

(B) 4.2

(C) 5.7

(D) 8.4

10.

MEDIUM · DESMOS

The table shows six data points. Which equation best fits the data?

x	1	2	3	4	5	6
y	14	19	21	28	31	36

(A) $y = 9.5x + 4.4$ (B) $y = 4.4x + 5$ (C) $y = 3.6x + 12$ (D) $y = 4.4x + 9.5$

11.

MEDIUM · NO CALCULATOR NEEDED

Residents of a town were surveyed about how they usually commute.

	Car	Bus	Bike
Under 30	36	14	10
30–49	44	16	12
50+	30	20	8

What proportion of the residents who do not usually commute by car are 50 or older? (Enter a decimal or a fraction.)

12.

MEDIUM · NO CALCULATOR NEEDED

A teacher compared two study methods. Of the 90 students who used Method A, 63 passed. Of the 60 students who used Method B, 42 passed. Which statement is supported by the data?

- (A) The pass rates for the two methods are the same. (B) Method B has the higher pass rate because it has fewer failures.
- (C) Method B has the higher pass rate because its group is smaller. (D) Method A has the higher pass rate because more students passed.

13.

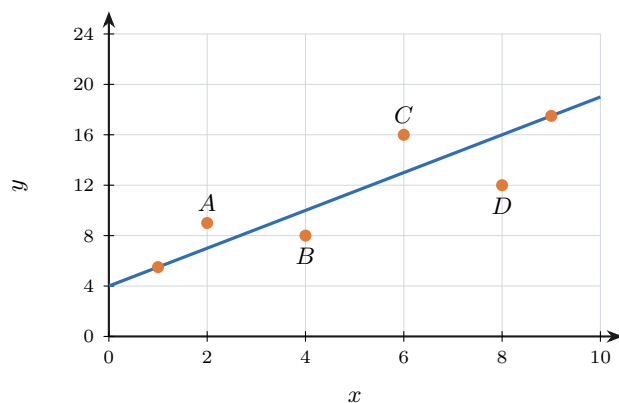
MEDIUM · NO CALCULATOR NEEDED

Each of 80 cards is numbered with a different integer from 1 to 80. One card is chosen at random. What is the probability that the number is a multiple of 6 and less than 50?

14.

HARD · NO CALCULATOR NEEDED

The scatterplot shows data and the line of best fit. Four of the data points are labeled A , B , C and D . For which labeled point is the absolute value of the residual greatest?

(A) Point C (B) Point D (C) Point A (D) Point B

15.

HARD · NO CALCULATOR NEEDED

A group of people chose between plans A and B .

	Plan A	Plan B
Member	24	x
Nonmember	36	28

The table shows the choices, where x is a positive integer. If one person who chose Plan B is selected at random, the probability that the person is a member is $\frac{3}{7}$. What is the value of x ?

16.

HARD · NO CALCULATOR NEEDED

The equation $\hat{y} = 12.5x + 140$ models the predicted number of visitors $\hat{y}$ to a museum each week, where x is the number of years since 2005. If t is defined as the number of years since 2010, which equation models the same relationship?

(A) $\hat{y} = 12.5t + 77.5$

(B) $\hat{y} = 62.5t + 140$

(C) $\hat{y} = 12.5t + 202.5$

(D) $\hat{y} = 12.5t + 140$

17.

HARD · NO CALCULATOR NEEDED

Runners trained either in a group or alone before a race.

	Finished	Did not finish
Group training	63	27
Solo training	36	54

Consider the following statements.

I. A greater proportion of group-training runners finished than of solo-training runners.

II. More than half of the runners who did not finish trained solo.

III. Fewer than 30% of the runners who finished trained in a group.

Which statements are true?

(A) I and II only

(B) II and III only

(C) I, II and III

(D) I only

18.

HARD · NO CALCULATOR NEEDED

Employees chose a retirement plan. The table shows the number choosing each plan by years of experience.

	< 5	5–9	10–14	15+
Plan A	20	24	20	12
Plan B	28	32	24	24

Of the employees with at least 10 years of experience, what proportion chose Plan B? (Enter a decimal or a fraction.)

19.

HARD · NO CALCULATOR NEEDED

A bag contains 5 red marbles and 7 blue marbles. Two marbles are drawn at random, one after the other, without replacement. What is the probability that both marbles are the same color?

(A) $\frac{37}{72}$
(C) $\frac{1}{2}$

(B) $\frac{5}{33}$
(D) $\frac{31}{66}$

20.

HARD · CALCULATOR

A random sample of 500 adults in a city of 24,000 adults was asked about age and cycling.

	Cycles	Does not cycle
Under 30	36	144
30 and over	44	276

Based on the sample, about how many adults in the city are under 30 and cycle?

21.

HARD · NO CALCULATOR NEEDED

Three training plans were compared.

	Plan A	Plan B	Plan C
Passed	30	26	22
Failed	20	14	18

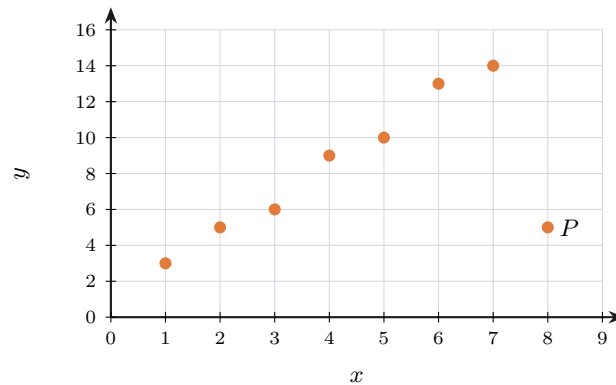
For which plan is the proportion of participants who passed equal to the proportion who passed among all three plans combined?

- (A) Plan C
- (B) All three are equal
- (C) Plan A
- (D) Plan B

22.

HARD · DESMOS

The scatterplot shows eight data points. A line of best fit is calculated for all eight points. If the point labeled P is removed and a new line of best fit is calculated, which describes the slope of the new line compared with the slope of the original line?



- (A) It cannot be determined. (B) It is greater.
(C) It is less. (D) It is the same.

23.

HARD · NO CALCULATOR NEEDED

A scatterplot has a mean x -value of 6 and a mean y -value of 31. The line of best fit has equation $\hat{y} = 2.4x + b$ and passes through the point (mean of x , mean of y). According to the line, what is the predicted value of y when $x = 10$?

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

A survey recorded whether students in grades 10 and 11 play a sport.

	Plays a sport	Does not play
Grade 10	24	36
Grade 11	30	k

The proportion of grade 10 students who play a sport equals the proportion of grade 11 students who play a sport. How many students were surveyed in all?

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

The table shows the relative frequencies, as fractions of all customers surveyed, for how customers shop and their age group. Some entries are missing.

	Under 40	40 or over	Total
Online	?	?	0.55
In-store	?	0.27	?
Total	0.50	?	1.00

Of the customers who are 40 or over, what fraction shop online?

(A) $\frac{23}{55}$
 (C) $\frac{32}{55}$

(B) $\frac{27}{50}$
 (D) $\frac{23}{50}$

26.

SAT CHALLENGE · CALCULATOR

In a town, 60% of the residents are adults and 40% are children. Of the adults, 35% own a bicycle, and of the children, 10% own a bicycle. A resident who owns a bicycle is selected at random. What is the probability that the resident is an adult? (Enter a decimal or a fraction.)

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

The least-squares line for five data points is $\hat{y} = 3x + 2$. The table shows four of the points.

x	1	2	3	4	5
y	7	5	12	?	?

Two y -values are missing, for $x = 4$ and $x = 5$. What is the missing y -value for $x = 5$?

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

A bag contains only red and blue marbles. The probability of randomly selecting a red marble from the bag is $\frac{2}{5}$. When 6 more red marbles are added to the bag, the probability of randomly selecting a red marble becomes $\frac{1}{2}$. How many blue marbles are in the bag?

(A) 18

(B) 24

(C) 30

(D) 12

SAT CHALLENGE • CALCULATOR

x	1	3
$f(x)$	24	54

(A) 206.25 (B) 198.25

(C) 182.25 (D) 108

SAT CHALLENGE • NO CALCULATOR NEEDED

SAT CHALLENGE · NO CALCULATOR NEEDED

Of the participants in the Business track, 35% attended online. How many of the 300 participants attended in person?

	Technical	Business	Total
In person	70		
Online	50		
Total	120		300

- (A) 113 (B) 145
(C) 187 (D) 63

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

A survey of 240 adults asked whether they commute by bicycle and whether they own a helmet. Of the adults who commute by bicycle, $\frac{3}{5}$ own a helmet. Of the adults who do not commute by bicycle, $\frac{1}{4}$ own a helmet. If one adult is selected at random from those who own a helmet, the probability that the selected adult commutes by bicycle is $\frac{12}{19}$. How many of the 240 adults own a helmet?

(A) 95

(B) 144

(C) 100

(D) 60